MAJOR PROJECT REGISTRATION FORM

Date:

Project Category:	☐ Research oriented ☐ Application oriented ☐ Product oriented ☐ Other (please specify) Specify if others:	Batch ID:	B5

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Title of the Project: AI-LLM Based Medical Imaging Classification and Segmentation

Abstract

The integration of Vision-Language Models (VLMs) into medical imaging represents a paradigm shift from purely pixel-based analysis to semantically grounded reasoning. Traditional deep learning approaches, while effective, often suffer from heavy reliance on large-scale annotated datasets and a lack of interpretability. This abstract explores the emerging application of VLMs— such as adaptations of CLIP, LLaVA, and domain-specific architectures like Med3DVLM—to the tasks of medical image classification and segmentation. By aligning visual features with clinical textual embeddings, VLMs enable zero-shot and few-shot learning capabilities, allowing models to generalize to unseen pathologies and modalities without extensive retraining.In classification, VLMs facilitate context-aware diagnosis and automated report generation by learning joint representations of radiological images and medical reports. In segmentation, the paradigm shifts toward "promptable" architectures (Vision-Language Segmentation Models or VLSMs), where text prompts (e.g., "segment the left ventricle") guide the model to delineate specific anatomical structures, significantly reducing the burden of manual annotation.

References (Category specific)

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2. **Badhan Kumar Das, Ajay Singh , Saahil Islam, Gengyan Zhao, (2017).** A SegResMamba: An Efficient Architecture for 3D Medical Image Segmentation.
   https://arxiv.org/html/2503.07766v1

3. **Ali Hatamizadeh, Vishwesh Nath, Yucheng Tang, Dong Yang, Holger Roth, Daguang Xu.**
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Project Guide

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